

SciQuery: Architecture & Smart Caching Explainer

NCERT Class 10 Science Doubt-Solving Assistant with Multi-Guard Semantic Cache

1. How the Chatbot Works

- Classification: Routes to STANDALONE, FOLLOW_UP, or STYLE using zero-LLM rules. First-turn follow-ups or style requests return helpful prompts with 0 LLM calls.
- Follow-Up Rewrite: Ambiguous follow-ups are rewritten to standalone questions via LLM; self-contained queries are preserved unchanged.
- Retrieval: Fetches top 6 chunks from 734 textbook chunks via FAISS using bge-small-en-v1.5 embeddings.
- Out-of-Scope Borderline Band: Score < 0.58 declines with 0 LLM calls; 0.58 to 0.68 lets the LLM inspect context; >= 0.68 answers directly.
- Generation: LLM answers strictly using context; numerals route to gpt-oss-120b with sign-convention prompts; failover on 429 to fallback model.

2. How the Cache Works

- Latency: Exact hit ~5 ms end-to-end (0.08 ms store); semantic hit ~11 ms (5.8 ms store); fresh LLM ~0.6s to 2s.
- Exact Match: Normalized queries (lowercased, punctuation-stripped) perform an instant SQLite lookup.
- Semantic Candidate Retrieval: Non-numerical queries search a FAISS vector index (cosine similarity >= 0.88).
- Five Safety Guards: Number Guard (identical values/units), Symbol Guard (exact match on letters C, F, P, 2F), Contrast Guard (concave/convex, acids/bases), Key-Term Guard (Jaccard >= 0.65), and Question-Type Guard (define vs law vs numerical).

3. What We Cache & What We Never Cache

- CACHED: Verified standalone NCERT answers with citations, canonical forms of follow-up questions, and contextual follow-up pairs.
- NEVER CACHED: Style restylings, out-of-scope declines, greetings, and numerical/calculation questions (containing digits or calculate/compute triggers) to prevent serving stale arithmetic.

4. What Did Not Work

- Lens Formula Sign Error Cached: A sign error in the lens formula (the model got 12 cm instead of 60 cm for $f = 20$ cm, $u = -30$ cm) was cached. Fix: sign-convention prompt, a stronger model for numerals (openai/gpt-oss-120b), and never cache numerals.
- Scope Threshold Calibration: With a 0.50 cutoff, off-topic questions that scored 0.60 to 0.67 (for example Mendeleev's periodic table at 0.6665 and renewable energy at 0.6741) got past retrieval. A single 0.68 cutoff then wrongly declined real questions ("What is a neuron?" at 0.6666, "What is saponification?" at 0.6158). So there are now three bands: below 0.58 decline without the LLM, 0.58 to 0.68 let the LLM decide, 0.68 and above answer.
- Pure Cosine Similarity: Opposing concepts scored deceptively high (concave vs convex: 0.9505; R=20 vs R=30: 0.9244). Required explicit contrast, symbol, and number guards to prevent factual hallucinations.
- Strict Jaccard Thresholds: A 0.80 Jaccard threshold caused false misses on natural phrasing variations; lowering to 0.65 preserved safety while allowing natural phrasing.

5. Architecture & Decision Flowchart

